

Detection and Classification of Human Emotion using Deep Learning Model

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Abstract— Human emotions are the mental states resulting from neurophysiological changes that are diversely linked to human thoughts, feelings, and behavioral reactions. Facial expressions, eye movement and gestures are several nonverbal communication methods used in many human-machine interactions. Facial emotion recognition has been an essential field because it has many applications that convey the mental state and emotions of each individual. As there is no clear way to distinguish between emotions on a face and the complicated neural system that is connected to the face, recognizing human emotions is a difficult undertaking. In this paper, for students' facial expression recognition, we presented a model called Convolutional Neural Network (CNN). The proposed model consists of six convolutional layers, six max pooling with two fully connected layers. Using a Haar cascade detector, the system identifies the faces of students from input images. It categorizes them into seven different facial expressions, including surprised, fearful, disgusted, sadness, happiness, anger, and neutral. The FER2013 dataset yielded an accuracy rate of 77.22% for the proposed model. The comparative study is carried out with four other deep learning models. The models are Inception V3, VGG-16, VGG-19, Resnet50 and the accuracies achieved are 40%, 37%, 22.50%, 22.09%, respectively.

Keywords— Emotion detection and classification, Deep learning Models, CNN.

Introduction

Human emotion detection is crucial in studying human-computer interfaces (HCIs), as emotions are elicited by particular circumstances [1]. As there is no clear way to distinguish between emotions on a face and the complicated neural system that is connected to the face, recognizing human emotions is a complex undertaking. This study uses convolutional neural networks to detect and classify human emotional expressions. Convolutional Neural Networks (CNN) have developed this work for human emotion expression recognition and classify them into seven basic categories: surprised, fearful, disgust, sadness, happiness, anger and neutral. Rather than computing features through hand-engineering, CNN uses automatic learning to calculate the features. By incorporating this novel method, the proposed model could achieve satisfied accuracy rates for recognizing human emotions. The proposed system deals with increasing accuracy with the help of various deep learning models for the vast dataset FER2013, which has 35587 greyscale images, including infant faces and male and female human faces. In this work, the accuracies carried out

for all the models are compared to obtain better results for the existing state of the art, and results are compared. The work compares the four other deep learning models-VGG-16, VGG-19, Inception V3, and Resnet50.

I. RELATED WORK

The papers stated below are studied in order to develop the most optimal and technically possible design, and the procedure is followed in accordance with the findings to reach a conclusion

The paper by Vishwesh Akre, Asif Nawaz, Pankaj Mohindru, and Sheeraz Ahmed examines a new functional trend that has been considered in emotion detection. Convolutional Neural Network (CNN) is the method proposed for the investigation. The proposed study compares the performance of two deep learning methods for emotion recognition: convolutional neural networks (CNN) and recurrent neural networks (RNN). The FER2013 dataset from Kaggle, which includes seven emotion classes, is used to train both neural networks [2].

The author Ved P Mishra has designed humans' main characteristic of behaviour that facilitates communication, which is their emotions. Deciphering nonverbal cues helps us understand spoken cues. Body language, gestures, postures of the hands, and other non-verbal cues are all part of non-verbal communication. Facial expressions account for around half of all non-verbal cues. Six primary emotions—sadness, happiness, anger, disgust, fear, and surprise have been distinguished from the myriad emotions that people experience. Humans can infer emotion and decipher facial motions on their own [3].

Bolla Gopi Krishna Reddy, Pitchuka Yashwanthsaai and Akurathi Ravi Raj has their task in a way that produces satisfactory results. The ability to perceive and identify emotions and facial expressions is a trait shared by all humans. The ability of computers to now carry out the same task brings many additional difficulties into daily life. Identifying offenders and providing specialised care are only a few examples of how emotion recognition can boost security, permit financial transactions without physical cards, and do other things [4].

In this paper, Akriti Jaiswal, A. Krishna Raju, and Suman Deb explain Human Emotion detection is one of the most influential and difficult social communication research tasks for emotion recognition from an image. Performance is improved Over conventional image processing approaches using deep learning (DL) based emotion identification. This study outlines the design of an artificial intelligence (AI) system that can identify emotions from a person's expressions on their face. The emotion detection process is covered, and it essentially consists of three steps: face detection, feature extraction, and emotion classification. This paper suggested a deep learning architecture for emotion identification from photos using CNN [5].

The author, Imane Lasri, investigates the idea of deep learning methods, which have been quite successful in several domains, including computer vision. A model for CNN may be trained to analyse images and recognise facial expression. In this research, we develop a system to identify students' emotions based on their facial expressions. Our approach entails three stages: face detection using Haar Cascades, normalisation, and emotion recognition using CNN on the FER2013 database with seven types of expressions. The findings indicate that facial emotion detection is practicable in the educational setting. As a result, it may assist teachers in adapting their presentations to the emotions of their pupils [6].

In this paper by Noel Jaymon and Sushma Nagdeote, the connection between emotions is expressed through facial expressions, which can reveal personality traits and mental processes. To expand its use in public, the machine continuously performs different activities. Many different duties can be carried out by machines that can comprehend emotions. Machine perception necessitates that machines be able to understand information about their surroundings. To comprehend emotions and find numerous applications in human-computer interaction, it is critical to recognise how people express themselves through their faces [7].

Prashant Dhope and Mahesh B. Neelagar have provided their research detailing the fundamental introduction. As artificial intelligence develops, machines are increasingly conscious of human emotions. The goal of the proposed method is to create a real-time facial expression recognition system using artificial intelligence. All seven basic emotions of the human face can be recognised using the suggested strategies. Anger, disgust, fear, joy, neutrality, sadness, and surprise are among them [8].

In this paper by Devanshu Shah Khushi Chavan, Sanket Shah, and Pratik Kanani, that humans' capacity for emotional expression is a key component of who we are. Communication between intimate partners is based on emotions. With a computer, it may also be possible to recognize emotions expressed through facial expressions, which might be highly useful. More intelligent technologies have used Facial emotion recognition in recent years to enhance human interaction. People's emotions influence these

systems constantly, changing how they operate. Here, we suggest a CNN-based architecture for identifying facial expressions of emotion [9].

As the authors, Lutfiah Zahara, Purnawarman Musa, and Eri Prasetyo Wibowo's paper discusses facial expressions as one of the ways people communicate. Deep learning techniques in human-computer interactions are used in research to advance artificial intelligence as an efficient system application procedure. One illustration would be if someone tried to communicate by using facial expressions and did so. Some people who see it occasionally fail to understand the expression or emotion it will elicit. In psychology, identifying emotions or facial expressions necessitates investigating and evaluating judgments in anticipating a person's or group's feelings during communication [10].

II. IMPLEMENTATION DETAILS

To implement the specified work described in the functional block diagram, the process is carried out using the Google Colab Platform

A. Tools and Platforms used

CNN Model is developed using Python with libraries, the OpenCV image processing implementation, Tensor Flow for convolutional neural network modelling, and Keras for machine learning. NumPy is used to carry out mathematical computations. The model is developed using Google Colab using the available GPUs.

B. Obtaining the Image Dataset

The dataset FER2013 is taken from Kaggle's facial expression recognition competition, which is used for emotion detection. The dataset is introduced by Aaron Courville and Pierre-Luc Carrier. His annotations are made-up of specific behaviours and feelings. The dataset has two sets of images: there are 28841 images in the training set and 7066 images in the validation set. The images are grayscale images of 48x48 in size. It is categorized into seven basic emotions: anger, disgusted, fearful, happiness, sadness, surprised, and neutral. The emotions collected are of people of different age groups, including infant faces, male and female human faces.

C. Image Augmentation Technique

Through various processing techniques, training images are intentionally created as part of the procedure used to enlarge the dataset artificially. A resilient convolution neural network that can classify objects even when they are positioned in multiple orientations. It keeps data from becoming scarce and gives models flexibility. Thanks to the target size, they can add to the images in real time while the model is still training. This both strengthens the model and spares it from consuming unnecessary memory. Gradient descent's batch size hyperparameter determines how much

training data must be processed before changing the model’s internal parameters.

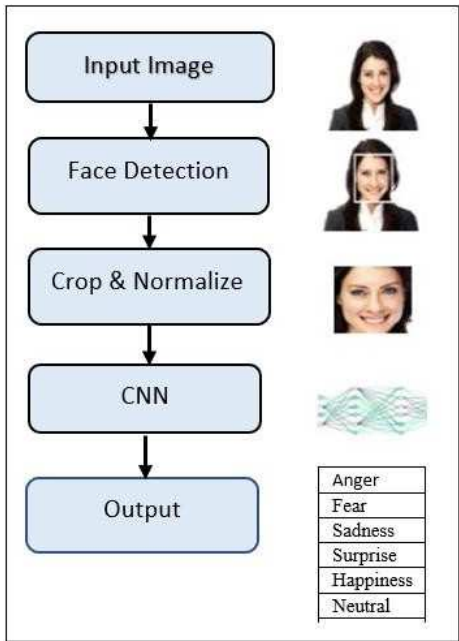


Fig. 1. Process flow of Emotion recognition

D. Layers and Activation functions used

- a. The Convolutional layer is the primary building block of a Convolutional network. It has a set of filters, parameters of which are to be learned throughout the training process. Sliding convolutional filters are applied to 2-D input by a 2-D convolutional layer. By moving the filters along the input vertically and horizontally, computing the dot product of the weights and the input, and adding a bias term, the layer convolves the input. Each filter convolves with the image and creates an activation map. Utilizing a weighted sum and bias addition, the activation function determines whether or not to stimulate a neuron. The study uses SoftMax activation function, which aims to introduce non-linearity and generates the output between ‘0’ and ‘1’. It determines each output, such that the total sum of the output is equal to ‘1’. SoftMax is often used for output layers and for multi-class classification.
- b. Dropout Layer: It is a regularization technique that overcomes the overfitting problem. In multilayered network, overfitting problem is common, so to resolve the problem, the regularization technique is used by removing the unwanted neurons for a temporary period during the training with neural network, it is a particularly effective model of model averaging.
- c. Dense Layer: Each neuron in the simple layer of neurons, known as the dense layer, receives information from every cell in the layer below it. Based on the results of the convolutional layers, a

dense layer is utilized to categories the images. Such neurons are found in large numbers inside a layer.

III. METHODOLOGY



Fig. 2. The CNN architecture

1. The preferred methodology can be split into two stages, that is model training and real-time detection. Fig. 1 shows the flow process of emotion recognition.
2. In this preferred approach, the goal is to analyze facial emotion recognition using CNN. This is the primary network and foundation for all the other deep learning models used in this experiment.
3. Initially, the image is fed as input to the model and the face is detected and cropped normalizing to the size of 48X48. Finally, emotion detection and classification are carried out accordingly.
4. The neuron is the key component of the CNN layer. They are interconnected so that the neurons at a given layer's output become the neurons at the subsequent layer's input. Back-propagation algorithm is used to compute the partial derivatives of the cost function. Convolution refers to using a filter or kernel on the input image to produce a feature map. The CNN model contains three types of layers, as shown in Fig. 2[6].
5. Pooling Layer: The pooling layer retains the critical feature in the image, which is fed as input and useful for classification. The different types of pooling are max pooling, average pooling and sum pooling. The pooling layer used in this study is max pooling. Max pooling takes the maximum value within the window (filter or kernel) and makes the layer resistant, which does not vary to small distortions or variations in the images.
6. Fully Connected layer: Fully connected implies each neuron at a given layer is interconnected to every other neuron in the successive layer. The input image is fed to the convolution and pooling layer, which extracts features based on the trained dataset. The output of this layer is sent to classify the emotions of all the images captured or video captured through a fully connected layer. OpenCV library is used to capture live frames from web cameras and to detect students faces based on Haar cascade method in a Keras, the ImageDataGenerator class is used to perform image augmentation. This class allowed us to transform the training images by rotation, shifts, shear, zoom and flip.

IV. EXPERIMENTAL RESULTS

The FER2013 dataset, which contains seven different emotions happiness, anger, sadness, disgust, neutral, fearful, and surprised was used to train the Convolutional Neural Network model. The identified facial image sizes were reduced to 48 by 48 pixels and transformed into grayscale pictures.



Fig. 3. Result of emotion classification of the image saved location

Human Emotion	Emotion value from the video
Anger	1.29
Disgust	0.00
Fearful	0.57
Happiness	197.79
Sadness	6.55
Surprised	1.75
Neutral	37.43

The CNN model then uses these as inputs. Two fully connected layers and six convolution and pooling layers make up the CNN model. In order to assess the effectiveness and quality, the suggested model has an accuracy rate of 77.2% at the 50 epochs. It has been noted that the model predicts both pleased and shocked emotions with efficiency. However, because it gets confused with sad expressions and the feared emotion, its prediction accuracy is low.

Table-1: Accuracies and loss of implemented model.

Model name	Accuracies and Loss achieved in %		
	Accuracy	Loss	Val Loss
CNN	77.22	1.0550	1.1734
VGG-16	37.71	1.0281	1.0254
VGG-19	22.50	1.6906	1.6267
InceptionV3	40	1.5206	1.5412
Resnet50	22.09	1.9574	2.0077

The comparative study, carried out for another four deep learning models are VGG-16, VGG-19, Inception V3 and Resnet50, with the achieved accuracies of 37%, 22.50%, 40%, and 22.09%, respectively. The above table gives the results regarding the study undertaken and verified results.

Model Training:

To enhance network parameters, batch size is set to 128 and epochs to 50 while training the model. The learning rate is

set to 10^{-3} . Categorical cross-entropy is chosen for the error function since it is helpful in classifying problems.

Testing in a real-time environment:

The trained model was assessed in a real-time setting following the development of the preferred CNN architecture. The model has been tested with both, the dataset and real-time pictures and videos captured from a webcam, and the results are as follows:

1. Image captured from webcam:



Fig.4. Result of emotion classification of the image captured from webcam.

Table 2: Result of emotion classification of the image captured from the webcam

Emotions	Emotion value from the image
Anger	0.04
Disgust	0.0
Fearful	0.02
Happiness	0.79
Sadness	0.04
Surprised	0.01
Neutral	0.1

2. Image captured from Dataset:



Fig.5. Result of emotion classification of the image from the dataset.

Table 3: Result of emotion classification of the image from the dataset

Emotions	Emotion value (from the image)
Anger	0.17
Disgust	0.01
Fearful	0.78
Happiness	0.01
Sadness	0.01
Surprised	0.01
Neutral	0.1

As demonstrated by the top two scores from Tables 2 and 3, live testing results validate the CNN model's high degree of confidence in identifying all emotions in real-time scenarios.

V. CONCLUSION AND FUTURE SCOPE

This article introduces a Convolutional Neural Network model for the facial expression recognition of the students. The preferred model comprises six convolutional layers and six max-pooling layers with two fully connected layers. Haar cascade detector is used to recognize students' faces as input images and classify them into different emotions such as surprised, fearful, disgust, sadness, happiness, anger and neutral. The preferred CNN model obtained an accuracy rate of 77.22% on the FER2013 dataset. To increase the accuracy percentage, a balanced set of images of all the expressions of emotion can be considered. There are significantly fewer images expressing the feeling of fear in this sample, indicating the dataset is imbalanced. So, having an equal number of images (balanced dataset) of all the emotions expressed in the dataset can be achieved as a part of the future scope. Further, it can also be implemented on 3D dataset images for greater percentage of accuracy. As a future scope, to make the model generalizable the model can be implemented with different dataset and compare the results.

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